# Notes: Keys, Pope, and Pope (JFE, 2016)

**Failure to Refinance**

## Contents

# 1 Big picture

- **Question.** How many U.S. households fail to refinance their fixed-rate mortgages when interest rates fall, even though refinancing would be financially beneficial?
- **Setting.** The paper studies a nationally representative sample of mortgage loans active in December 2010, a moment of unusually low mortgage rates after the Great Recession.
- **Core puzzle.** Refinancing can save households tens of thousands of dollars, yet many borrowers continue to make payments on mortgage rates well above the market rate.
- **Main quantitative result.** After imposing conservative restrictions designed to isolate borrowers who likely *could* refinance, the paper estimates that about **20%** of households for whom refinancing was optimal had not done so by December 2010.
- **Magnitude of the mistake.** For these households, the median forgone gain is about **$160 per month**, or roughly **$45,500** in unadjusted lifetime interest savings and **$11,500** in present-discounted, tax-adjusted savings.
- **Persistence.** This is not a purely short-lived delay. Even after mortgage rates fell further through 2012, about **40%** of the households identified as failing to refinance in December 2010 were still in the same home and still had not refinanced.
- **Micro evidence.** A nonprofit lender's mail campaigns to preapproved, eligible borrowers with no up-front out-of-pocket refinancing costs still generated take-up rates of only about **13%–24%**. This strongly suggests that the anomaly is not just a data artifact.
- **Behavioral interpretation.** The paper does not claim a single mechanism, but the evidence points toward procrastination, inattention, limited understanding, and other behavioral barriers rather than pure inability to refinance.
- **Macro implication.** Low policy rates do not automatically pass through to household debt-service relief if many borrowers fail to refinance. This weakens monetary transmission and may also limit default prevention.
- **Why the paper matters.** This is a clean household-finance paper about a very large financial mistake. It combines nationally representative loan-level evidence with direct lender-side evidence and connects individual behavior to macroeconomic policy.

# 2 Data and measurement (key objects)

## 2.1 Observed loan-level panel from CoreLogic

- The main dataset is a random sample of approximately **one million** mortgage loans that were active in **December 2010**.
- The data come from **CoreLogic Solutions** through a CoreLogic Academic Research Council data grant.
- Mortgage-level records are provided by most of the top 20 servicers in the nation and span both the agency and non-agency segments of the mortgage market.

- The overall CoreLogic database covers roughly **85%** of the U.S. mortgage market.
- To make households comparable, the sample is restricted to:
  - fixed-rate mortgages,
  - single-family owner-occupied homes,
  - non-FHA/VA loans,
  - non-manufactured and non-mobile homes,
  - loans not in foreclosure as of December 2010,
  - outstanding balances of at least $75,000.

## 2.2 Observed variables on each mortgage

- Date of origination.
- Borrower FICO score at origination.
- Loan-to-value ratio (LTV) at origination.
- Unpaid balance in December 2010.
- Contract interest rate.
- Time remaining on the loan.
- Zip code of the property.
- Full mortgage payment history, including missed and late payments.
- Information on all additional mortgage liens held by the household.

## 2.3 Merged geographic controls

- Zip-code Census data for median income and education.
- County unemployment rates from the Bureau of Labor Statistics.
- Zip-code housing price data from Zillow, used to update LTV and CLTV ratios through time.

## 2.4 Sample restriction ladder for likely refinancing eligibility

The paper progressively narrows the sample to approximate households who were likely both able and eligible to refinance in December 2010.

- Full sample: 994,188 loans.
- Add initial underwriting screen FICO $>$ 680 and initial LTV $<$ 90: 650,490 loans.
- Restrict to borrowers who *never missed a payment*: 573,973 loans.
- Restrict to current LTV $<$ 90: 477,601 loans.

- Restrict to current CLTV < 90: 376,036 loans.

This last sample is the paper's preferred benchmark for measuring the failure to refinance.

## 2.5 Key measured objects

- **Positive unadjusted savings.** Whether refinancing at the prevailing market rate would reduce total interest payments over the remaining life of the mortgage, net of assumed transaction costs.
- **Optimal refinancing in December 2010.** Whether refinancing passes the Agarwal–Driscoll–Laibson (2013) optimal threshold rule, which accounts for tax incentives, discounting, mobility, up-front costs, and the option value of waiting.
- **Adjusted savings.** Present-discounted savings that account for taxes, discounting, and the probability of moving.

## 2.6 Unobservables and timing

- The paper does *not* observe updated borrower income or credit score in the main CoreLogic sample at the December 2010 date.
- The paper observes *prepayments*, not refinancing directly, so later prepayment can reflect either refinancing or moving.
- The paper also does not directly observe prepayment penalties, though these are likely small or expired for most relevant fixed-rate mortgages by 2010.
- Household expectations about future rates, patience, and moving decisions are not observed; these enter through the optimal-threshold calculation.

## 2.7 Micro-level nonprofit lender sample

The paper complements the national sample with administrative data from **Neighborhood Housing Services of Chicago (NHS)** and its lending affiliate **Neighborhood Lending Services (NLS)**.

- These data identify households that were preapproved to refinance.
- The lender knew these households, serviced their loans, and screened them for refinancing eligibility.
- In the first campaign, appraisal and origination fees could be rolled into the new loan, so no cash up front was required.
- This micro setting is valuable because it isolates borrowers who were specifically identified by a lender as both eligible and likely to benefit from refinancing.

# 3 Models and empirical specifications (all equations + interpretation)

## 3.1 Mortgage refinancing as an intertemporal choice

The relevant comparison is between keeping the current fixed-rate mortgage and replacing it with a new one at the prevailing market rate.

- A standard fixed-rate mortgage (FRM) gives a constant nominal mortgage rate over the life of the loan.
- Refinancing means prepaying the old mortgage and originating a new one at the lower prevailing rate.
- A household should refinance when the expected discounted savings from lower future debt-service payments exceed the current transaction costs and the value of keeping the option to refinance later.

## 3.2 A convenient representation of the paper's savings calculation

The paper does not write the accounting exercise in this exact notation, but its empirical calculation is equivalent to the following.

Let

$$M(r, B, T) = B\frac{i(1+i)^T}{(1+i)^T - 1}, \qquad i = \frac{r}{12},$$

be the monthly payment on a mortgage with annual rate $r$, unpaid balance $B$, and remaining term $T$ months.

Then remaining total interest on the outstanding mortgage is

$$I(r, B, T) = T\, M(r, B, T) - B.$$

If the current contract rate is $r_i$ and the prevailing market rate is $r_t^m$, a natural measure of lifetime refinancing gains is

$$S_i^{\text{gross}} = I(r_i, B_i, T_i) - I(r_t^m, B_i, T_i) - \kappa_i, \tag{1}$$

where the assumed transaction cost is

$$\kappa_i = 0.01B_i + 2000. \tag{2}$$

- **Interpretation.** Equation (1) is the paper's basic unadjusted savings measure.
- **Immediate implication.** Large loan balances create mechanically larger gains from refinancing because the fixed cost is spread over a larger debt base.

## 3.3 Adjusted present-discounted savings

The paper then moves from a simple accounting calculation to an economically meaningful present value that adjusts for taxes, discounting, and the probability that the household moves before realizing all future savings.

A compact way to represent the paper's adjusted calculation is

$$S_i^{\text{adj}} = \sum_{t=1}^{T_i} \frac{\Pr(\text{still in home at } t)}{(1+\rho)^t} \Big[(1-\tau)\,\Delta M_{it}\Big] - \kappa_i, \tag{3}$$

where $\rho$ is the discount rate, $\tau$ is the marginal tax rate, and $\Delta M_{it}$ is the reduction in monthly mortgage outlays from refinancing.

- **Interpretation.** This equation explains why the present-discounted gains are much smaller than the raw lifetime interest savings.
- **Economic point.** The household may move before realizing the long-run benefit, mortgage interest is tax deductible, and future savings are discounted.

### 3.4 Optimal refinancing threshold from Agarwal, Driscoll, and Laibson (2013)

Rather than labeling every household with positive gross savings as someone who *should* refinance, the paper uses the optimal closed-form rule from Agarwal, Driscoll, and Laibson (2013), implemented through their "square-root rule" approximation.

The decision rule can be summarized as

$$\text{Refinance if } r_i - r_t^m \geq \Delta r_i^*(\tau, \rho, \mu, \kappa_i, B_i, T_i), \tag{4}$$

where:

- $r_i$ is the current contract rate,
- $r_t^m$ is the current market mortgage rate,
- $\tau$ is the marginal tax rate,
- $\rho$ is the discount rate,
- $\mu$ summarizes mobility / moving risk,
- $\kappa_i$ is the up-front refinancing cost.

The paper uses conservative calibration values from Agarwal, Driscoll, and Laibson:

$$\rho = 0.05, \qquad \tau = 0.28, \qquad \Pr(\text{move in a year}) = 0.10. \tag{5}$$

- **Interpretation.** A borrower should refinance only if the contract-market rate spread is large enough to cover both the direct costs of refinancing and the option value of waiting.
- **Practical implication.** For plausible parameter values, refinancing generally becomes optimal only after rates fall by about **100–200 basis points**.
- **Sensitivity.** The threshold is especially sensitive to up-front costs: the paper notes that every additional $1,000 of up-front cost moves the optimal threshold by roughly **25 basis points**.

### 3.5 Empirical implementation in December 2010

The prevailing market rate is taken to be the Freddie Mac PMMS average rate for a 30-year fixed-rate mortgage in November 2010:

$$r_t^m = 4.3\%. \tag{6}$$

For each loan, the paper computes:

- whether $S_i^{\text{gross}} > 0$, and
- whether the refinancing rate gap exceeds the optimal threshold in (4).

This generates the two main columns in Table 2:

- **Share with positive unadjusted savings**, and
- **Percent optimal in December 2010**.

### 3.6 Main national-sample results

The broad pattern is:

- In the full sample, **91.4%** of households have positive gross savings from refinancing, but only **41.2%** satisfy the optimal refinancing rule.
- Once the sample is restricted to likely-eligible borrowers, the preferred estimate falls to **20.0%**.

In the preferred sample, the typical forgone gain is large:

$$\text{Median gross gain} = \$45{,}473, \tag{7}$$

$$\text{Median adjusted present value gain} = \$11{,}568. \tag{8}$$

The paper interprets these as about **$160 per month** in forgone payment savings for the median household in the failure-to-refinance group.

### 3.7 Robustness and validation of eligibility assumptions

A key concern is that some borrowers in the "should refinance" group may actually have become ineligible because their credit deteriorated after origination.

The paper addresses this with an auxiliary credit-score exercise using a separate Equifax/CRISM-style sample that closely approximates the restricted CoreLogic sample.

- Among borrowers who originated with good credit, low LTV, and never missed a payment, **90.7%** still had FICO scores above 680 in December 2010.
- This suggests the main estimate is not driven by widespread credit-score deterioration.
- If the remaining **9.3%** are treated as actually ineligible, the failure-to-refinance estimate falls from **20.0%** to about **18.1%**.

The paper also varies the threshold assumptions.

- A naive **150 basis point** rate-gap rule gives a result broadly similar to the baseline, suggesting about **16%** should refinance and would save about **$140 per month**.
- Results are not especially sensitive to alternative patience or moving parameters.
- Results are very sensitive to up-front costs: with *no* up-front costs, the paper estimates that **94.6%** of households would have found refinancing optimal.

### 3.8 Heterogeneity analysis

The preferred sample is then split by borrower, mortgage, and geographic characteristics.

- **Unemployment.** Failure-to-refinance rates are remarkably similar across county unemployment quartiles: about **19.0%** in the lowest unemployment quartile and **20.2%** in the highest.
- **FICO.** There is a strong gradient by credit score even within the restricted sample: **29.1%** in the lowest FICO quartile versus **12.3%** in the highest.
- **CLTV.** There is also a gradient by current leverage: **17.5%** in the lowest CLTV quartile versus **23.4%** in the highest.
- **Loan amount.** The failure rate is surprisingly flat across balance size: roughly **21.2%** in the smallest-balance quartile and **19.1%** in the largest-balance quartile, even though potential savings rise massively with loan size.
- **Income and education.** Zip-code income and education show only limited gradients.
- **Interpretation.** The anomaly is not confined to distressed labor markets or low-income areas. It appears broad-based.
- **But.** It is more severe among borrowers with weaker credit histories or higher leverage, which is consistent with either tighter refinancing frictions or lower financial sophistication.

### 3.9 Geographic and market-structure variation

The paper computes state-level and MSA-level refinancing failures.

- State-level failure-to-refinance rates range from roughly **8%–9%** in places such as Mississippi, Wisconsin, and Minnesota to roughly **30%** in Florida, Oklahoma, and New York.
- There is no obvious regional clustering.
- At the MSA level, the share of borrowers failing to refinance is *negatively*, not positively, related to lender concentration as measured by the top-four-lender share.
- **Interpretation.** The paper does not find evidence that simple mortgage-market concentration is the main driver of the refinancing anomaly.
- **Caution.** This is descriptive, not causal.

### 3.10 Micro-level letter campaigns

The nonprofit lender evidence is organized around three waves of mail campaigns.

**Wave 1.** In July 2011, NLS sent letters to **446** preapproved borrowers offering a refinance rate of **4.7%**. No cash up front was required because fees could be rolled into the new loan.

**Wave 2.** In July 2012, NLS sent letters to **140** borrowers at **3.99%**.

**Wave 3.** In May 2013, NLS sent letters to **193** borrowers at **4.0%**. The paper randomized the wording of the letters, including one version that highlighted savings more directly.

### 3.11 Reduced-form take-up evidence from the micro sample

The take-up rates remain low despite preapproval and large financial gains:

- Wave 1: **15.9%** refinance.
- Wave 2: **24.3%** refinance.
- Wave 3: **13.0%** refinance.

Median unadjusted savings for those who refinanced were large:

- Wave 1: $24,500.
- Wave 2: $29,900.
- Wave 3: $48,200.

But non-refinancers also left a lot of money on the table:

- Wave 1: $17,700.
- Wave 2: $24,700.
- Wave 3: $26,400.

Pooling across waves, the paper estimates simple reduced-form predictors of take-up. The only robust predictor is the loan amount: an extra $10,000 in loan size raises take-up by about **1.2 percentage points** on a baseline refinancing rate of **16.7%**. Conditional on loan amount, the origination rate is not significantly related to take-up.

### 3.12 Survey evidence on mechanisms

After Wave 3, the authors surveyed non-refinancers.

- Of the households reached, up to about **one-quarter** said they did not open the letter.
- Of those who opened it, a little more than **one-third** said they planned to call but never got around to it or were too busy.

- Another **one-third** said they did not think the savings were large enough.
- At the end of the survey, **12 out of 32** respondents said they would be happy to have a loan officer call them to discuss refinancing.

- **Interpretation.** These answers line up well with procrastination, inattention, misunderstanding, and perhaps limited trust or limited salience.
- **Importantly.** They are much harder to reconcile with a pure liquidity-constraint story because the offers were preapproved and involved no required cash up front.

# 4 Identification and instruments (what makes the estimates credible)

## 4.1 What fails in a naive interpretation

A naive comparison of current mortgage rates to market rates would overstate the anomaly.

- Some households are not eligible to refinance because their credit deteriorated.
- Some are too underwater on their mortgage or on their combined liens.
- Some later prepayments reflect moving rather than refinancing.
- Some households may rationally delay because refinancing has option value if rates may fall further.

This is why the paper does *not* simply count all high-rate borrowers as making mistakes.

## 4.2 What identifies the national-sample anomaly

The paper identifies a meaningful failure to refinance through a sequence of conservative design choices.

- It restricts the sample to borrowers with strong initial underwriting characteristics.
- It uses updated current LTV and CLTV measures from house-price data.
- It requires pristine payment history, which serves as a proxy for continued creditworthiness.
- It applies the Agarwal–Driscoll–Laibson optimal rule rather than a crude positive-savings rule.
- It validates the credit-score proxy with auxiliary data showing that 90.7% of the relevant borrowers still have FICO above 680 in December 2010.

- **Interpretation.** The paper cannot prove that every one of the 20% could refinance, but it shows that the anomaly survives under an intentionally conservative screen.

## 4.3 Why the micro evidence is so valuable

The micro evidence sharpens identification because it strips away several obvious alternative explanations.

- Borrowers are preapproved by the lender.
- The lender itself believes they are eligible and would benefit.
- Up-front out-of-pocket cost is not required.
- The offer comes from a lender with an existing relationship to the borrower.

Low take-up in this setting is hard to explain purely by institutional ineligibility.

### 4.4 Limits of identification

The paper is very convincing about the *existence* of the anomaly, but more limited about its exact mechanism.

- It does not causally separate inattention, procrastination, low trust, financial illiteracy, or misunderstanding.
- It cannot fully distinguish refinancing from moving in the later prepayment exercise.
- The nonprofit lender sample is not nationally representative; it is disproportionately concentrated in lower-income Chicago neighborhoods.
- The randomized letter variation in Wave 3 is underpowered, so the paper cannot say much about which exact framing interventions work best.

### 4.5 Relation to the literature

- It builds on mortgage-refinancing theory and evidence, especially Agarwal, Rosen, and Yao (2012) and Agarwal, Driscoll, and Laibson (2013).
- It connects to the broader literature on costly household financial mistakes and limited take-up of beneficial programs.
- It also speaks directly to post-crisis policy debates around HARP, automatic refinancing, and the pass-through of monetary policy to households.

## 5 Tables and figures

### 5.1 Table 1 and Figure 1: Sample composition and the rate distribution

**Read:**

- The unrestricted sample is large and representative, but the paper's main analysis deliberately moves to tighter and tighter screens to avoid overstating the anomaly.
- The preferred sample still has 376,036 mortgages, so the 20% estimate is not coming from a tiny corner of the data.
- Even among likely-eligible borrowers, the distribution of contract rates remains wide and many borrowers are still paying rates far above the 4.3% market rate in December 2010.

**Table 1**
This table provides summary statistics (average values) for the CoreLogic sample of mortgage loans originated prior to November 2010 and active in December 2010. Each column represents a tighter underwriting standard for potential refinancing by lenders. See text for detailed description of sample selection criteria. LTV = loan to value.

| | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| Variables of interest | | | | | |
| Interest rate | 5.52 | 5.29 | 5.22 | 5.22 | 5.10 |
| Years remaining | 23.4 | 23.3 | 23.3 | 22.9 | 22.9 |
| Unpaid balance | $205,218 | $215,481 | $215,248 | $216,296 | $212,102 |
| Monthly payment | $1,370 | $1,421 | $1,414 | $1,420 | $1,395 |
| FICO score at origination | 737 | 758 | 761 | 761 | 765 |
| LTV at origination | 70.7 | 66.4 | 65.9 | 64.6 | 62.7 |
| Computed LTV in December 2010 | 74.2 | 68.5 | 67.1 | 62.4 | 60.2 |
| Sample restrictions | | | | | |
| FICO > 680 and LTV < 90 | No | Yes | Yes | Yes | Yes |
| Never missed a payment | No | No | Yes | Yes | Yes |
| Current LTV < 90 | No | No | No | Yes | Yes |
| Current CLTV < 90 | No | No | No | No | Yes |
| Number of observations | 994,188 | 650,490 | 573,973 | 477,601 | 376,036 |

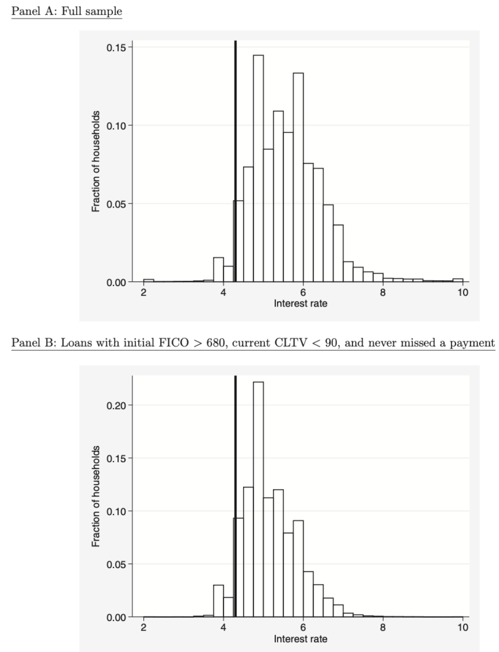


**Fig. 1.** Distribution of interest rates in December 2010. Panel A shows the distribution of interest rates of mortgage loans originated prior to November 2010 and active in December 2010. Panel B shows the distribution of interest rates for a restricted sample of loans with a high likelihood of refinancing eligibility: FICO scores above 680, current combined loan-to-value (CLTV) ratio below 90%, and never missed a payment. The vertical line indicates the prevailing mortgage rate of 4.3% in December of 2010. See text for detailed description of sample selection criteria.

- The key visual message of Figure 1 is that the refinancing option was economically relevant for a large mass of households, not a tail event.

## 5.2 Table 2: Main estimate of the scope and size of failure to refinance

**Read:**

- The difference between **positive savings** and **optimal refinancing** is important. Nearly everyone can save something mechanically, but far fewer should refinance once discounting, taxes, mobility, and option value are recognized.
- The paper's headline result is the preferred-sample row: **20.0%** should optimally refinance but have not done so.
- The median forgone adjusted gain of **$11,568** shows this is not a small mistake.
- The raw economic stakes are even larger: the median unadjusted gain is **$45,473**.

## 5.3 Figure 2: Distribution of forgone savings

**Read:**

- There is substantial heterogeneity in the dollar magnitude of the mistake.
- The median household in the failure-to-refinance group leaves about $45,000 of unadjusted savings on the table.

**Table 2**
This table provides estimates of the fraction of mortgage borrowers who optimally could have refinanced in December of 2010 but had not done so and the savings had they refinanced. The sample consists of mortgage loans originated prior to November 2010 and active in December 2010. See text for detailed description of sample selection criteria and savings calculations. All savings calculations include transaction costs of one point (1% of the unpaid balance) plus $2,000. Optimal threshold is calculated using Agarwal, Driscoll, and Laibson (2013) formula.

| Sample | Number of observations | Share with positive unadjusted savings | Percent optimal in December 2010 | Median unadjusted savings if optimal | Median adjusted savings if optimal |
|---|---|---|---|---|---|
| Full sample | 994,188 | 91.4 | 41.2 | $54,313 | $13,260 |
| Initial FICO > 680 and initial LTV < 90 | 650,490 | 89.0 | 31.1 | $53,831 | $13,218 |
| Initial FICO > 680 and initial LTV < 90, never missed a payment | 573,973 | 88.2 | 27.5 | $52,075 | $12,815 |
| Initial FICO > 680 and current LTV < 90, never missed a payment | 477,601 | 87.2 | 23.4 | $48,344 | $12,174 |
| Initial FICO > 680 and current CLTV < 90, never missed a payment | 376,036 | 85.7 | 20.0 | $45,473 | $11,568 |

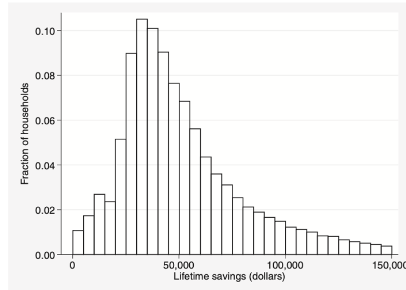


**Fig. 2.** Distribution of unadjusted lifetime savings for the 20% of households that should optimally refinance in December 2010. Savings are calculated based on the remaining unpaid balance, the remaining loan term, and the difference between the market interest rate and the interest rate at origination. The sample consists of mortgage loans originated prior to November 2010 and active in December 2010 with a high likelihood of refinancing eligibility: FICO scores above 680, current combined loan-to-value (CLTV) ratio below 90%, and never missed a payment. See text for detailed description of sample selection criteria.

- The upper tail is large: one-quarter would save more than roughly $68,000 and about 11% more than $100,000.
- The anomaly is therefore not only widespread but also economically large for many households.

## 5.4 Table 3: Who fails to refinance?

**Read:**

- The anomaly is broad-based. It does not disappear in low-unemployment counties, high-income areas, or more educated zip codes.
- There is a strong gradient by credit quality and leverage: low-FICO and high-CLTV households fail to refinance more often.
- Strikingly, loan size changes the *stakes* enormously but barely changes the *failure rate*. Households with the biggest dollar gains are not much more likely to refinance.
- This last fact is highly suggestive of behavioral frictions: people do not appear very responsive to very large differences in the monetary benefit.

**Table 3**
This table provides estimates of the fraction of mortgage borrowers who optimally could have refinanced in December of 2010 but had not done so, and the savings had they refinanced, across a range of borrower, mortgage, and geographic characteristics. The sample consists of mortgage loans originated prior to November 2010 and active in December 2010. See text for detailed description of sample selection criteria and savings calculations. All savings calculations include transaction costs of one point (1% of the unpaid balance) plus $2,000. Optimal threshold is calculated using Agarwal, Driscoll, and Laibson (2013) formula.

| | Share with positive unadjusted savings | Percent optimal in December 2010 | Median unadjusted savings if optimal | Quartile range |
|---|---|---|---|---|
| By unemployment rate quartile | | | | |
| Least unemployment | 84.5 | 19.0 | $47,667 | < 7.7% |
| Second quartile | 85.3 | 20.6 | $44,432 | 7.7% < x < 9.2% |
| Third quartile | 86.3 | 20.0 | $44,561 | 9.2% < x < 10.9% |
| Most unemployment | 86.8 | 20.2 | $45,520 | > 10.9% |
| By FICO score quartile | | | | |
| Lowest FICO | 90.0 | 29.1 | $46,150 | < 741 (but > 680) |
| Second quartile | 87.3 | 21.3 | $45,651 | 741 < x < 773 |
| Third quartile | 84.3 | 16.4 | $45,307 | 773 < x < 793 |
| Highest FICO | 81.1 | 12.3 | $43,670 | > 793 |
| By current CLTV quartile | | | | |
| Lowest CLTV | 79.1 | 17.5 | $37,285 | < 54% |
| Second quartile | 83.9 | 18.9 | $42,518 | 54% < x < 69% |
| Third quartile | 87.7 | 19.6 | $47,429 | 69% < x < 80% |
| Highest CLTV | 91.4 | 23.4 | $52,437 | > 80% (but < 90%) |
| By loan amount quartile | | | | |
| Smallest loan amount | 85.4 | 21.2 | $30,324 | < $140,000 |
| Second quartile | 83.0 | 20.6 | $39,496 | $140,000 < x < $196,000 |
| Third quartile | 85.0 | 19.0 | $53,248 | $196,000 < x < $288,000 |
| largest loan amount | 89.5 | 19.1 | $94,599 | > $288,000 |
| By education quartile | | | | |
| Least educated county | 85.4 | 19.6 | $39,846 | < 28.7% |
| Second quartile | 86.4 | 22.1 | $45,488 | 28.7% < x < 33.8% |
| Third quartile | 85.4 | 18.8 | $46,033 | 33.8% < x < 41.1% |
| Most educated county | 85.6 | 19.1 | $53,292 | > 41.1% |
| By Income quartile | | | | |
| Least income county | 85.4 | 20.7 | $40,467 | < $53,418 |
| Second quartile | 85.8 | 19.8 | $43,840 | $53,418 < x < $61,555 |
| Third quartile | 85.6 | 18.8 | $46,618 | $61,555 < x < $75,566 |
| Most income county | 86.0 | 20.4 | $52,663 | > $75,566 |

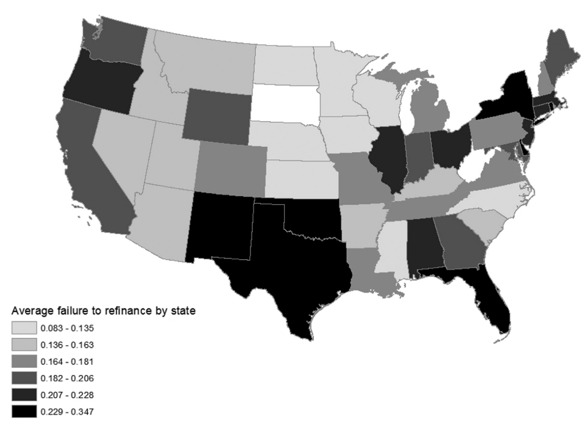


**Fig. 3.** Geographic variation in the failure to refinance. The figure shows state-level failure-to-refinance rates based on households that should optimally refinance in December 2010. The sample consists of mortgage loans originated prior to November 2010 and active in December 2010 with a high likelihood of refinancing eligibility: FICO scores above 680, current combined loan-to-value (CLTV) ratio below 90%, and never missed a payment. See text for detailed description of sample selection criteria.

## 5.5 Figure 3: Geographic variation by state

**Read:**

- Failure-to-refinance rates vary a lot across states, from single digits in some places to around 30% in others.
- But there is no obvious regional pattern that would point to one simple institutional explanation.
- The paper therefore interprets geography as suggestive rather than definitive evidence on mechanisms.

## 5.6 Figure 4: Failure to refinance and lender concentration

**Read:**

- The correlation between the failure-to-refinance rate and local top-four lender share is slightly negative.
- This goes against the simple idea that more concentrated mortgage markets mechanically generate less refinancing.
- The refinancing anomaly therefore seems unlikely to be explained mainly by straightforward lender concentration stories.

**Table 4**
This table summarizes the three waves of Neighborhood Lending Services (NLS) refinancing mail campaigns, undertaken in May 2011 (Wave 1), July 2012 (Wave 2), and May 2013 (Wave 3). The first two waves included outgoing calls from loan officers. The third wave was exclusively conducted by mail.

| Mail campaign characteristics, by wave | |
|---|---|
| Wave 1 | |
| Number of letters sent | 446 |
| Percent that refinanced | 15.9 |
| Median original interest rate | 6.2% |
| Median unadjusted savings for those that refinanced | $24,500 |
| Median unadjusted savings for those that did not refinance | $17,700 |
| Wave 2 | |
| Number of letters sent | 140 |
| Percent that refinanced | 24.3 |
| Median original interest rate | 6.1% |
| Median unadjusted savings for those that refinanced | $29,900 |
| Median unadjusted savings for those that did not refinance | $24,700 |
| Wave 3 | |
| Number of letters sent | 193 |
| Percent that refinanced | 13.0 |
| Median original interest rate | 6.1% |
| Median unadjusted savings for those that refinanced | $48,200 |
| Median unadjusted savings for those that did not refinance | $26,400 |

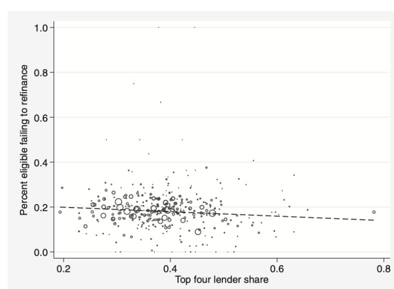


**Fig. 4.** Metropolitan Statistical Area (MSA)–level relationship between refinancing and lender concentration. The figure shows the correlation between the share of households that should optimally refinance in December 2010 but failed to do so in an MSA and the degree of lender concentration in that MSA. Lender concentration is measured as the share of the top 4 lenders in the MSA. Circle size represents the number of observations in our sample from the MSA. The sample consists of mortgage loans originated prior to November 2010 and active in December 2010 with a high likelihood of refinancing eligibility: FICO scores above 680, current combined loan-to-value (CLTV) ratio below 90%, and never missed a payment. See text for detailed description of sample selection criteria.

## 5.7 Table 4: Micro letter campaigns

**Read:**

- The key fact is the very low take-up despite preapproval, trusted-lender contact, and no required up-front cash.

- The anomaly survives in an environment where the lender is actively trying to help the borrower refinance.
- Refinancers do have slightly larger potential gains, but non-refinancers also pass up very large savings.
- This is the paper's cleanest evidence that the failure to refinance is a real household decision problem rather than only an artifact of national administrative data.

### 5.8 Robustness and extensions

**Read:**

- The robustness analysis is important because the paper is fundamentally about distinguishing true ineligibility from non-optimal inaction.
- The credit-score validation and alternative-threshold exercises show that the exact percentage moves around, but the qualitative conclusion does not: many households still fail to refinance when it is privately beneficial to do so.
- The anomaly becomes much smaller only when one assumes much larger refinancing frictions than the paper's conservative baseline or when one removes the possibility of eligibility very aggressively.

## 6 Short summary

- **Conclusion.** A substantial fraction of U.S. mortgage borrowers fail to refinance even when it is optimal and apparently feasible for them to do so.
- **Core evidence.** In the preferred national sample, about 20% fail to refinance; in the nonprofit lender sample, most preapproved borrowers still ignore direct offers that would lower their payments.
- **Main lesson.** Household financial frictions are not just about market access. Even when access is plausibly available, behavioral barriers can prevent households from taking large gains.

## 7 Conclusion

### 7.1 Core conclusion (what the paper establishes)

The paper establishes that the U.S. refinancing margin is far from frictionless. During the low-rate environment after the Great Recession, a nontrivial set of households continued paying rates that were clearly too high relative to the available market rate. This is true even after imposing conservative screens designed to isolate borrowers who likely remained eligible to refinance. The paper therefore identifies a large and economically meaningful household financial mistake.

## 7.2 Economic intuition

Refinancing looks simple in theory but is hard in practice. The gains arrive slowly over time, while the hassle and direct costs are immediate. Borrowers must understand a complex optimization problem, trust the lender, gather documentation, and act. Even modest procrastination or misunderstanding can therefore generate a large wedge between what is optimal and what households actually do. The nonprofit lender evidence makes this especially clear: even when a lender says, in effect, "you are eligible and this saves you money," most borrowers still do not respond.

## 7.3 Takeaway

The big message of the paper is that lowering market interest rates is not enough to guarantee lower household debt-service costs. Monetary policy, refinancing programs like HARP, and household balance-sheet repair all depend on borrowers actually acting on the refinancing option. If many do not, then rate cuts pass through incompletely, households leave large sums on the table, and macro stimulus is weakened. This is why the paper ultimately points toward policy ideas such as streamlined refinancing or automatically refinancing mortgages: the friction is not only in market prices, but in household action itself.